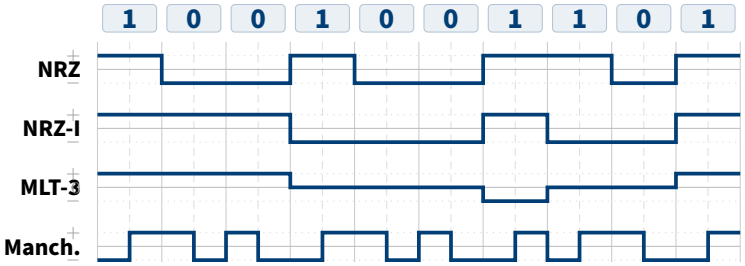
Aufgabe 1: Bitcodierungen

Leitungscodierung & Signalverläufe – Übungsblatt

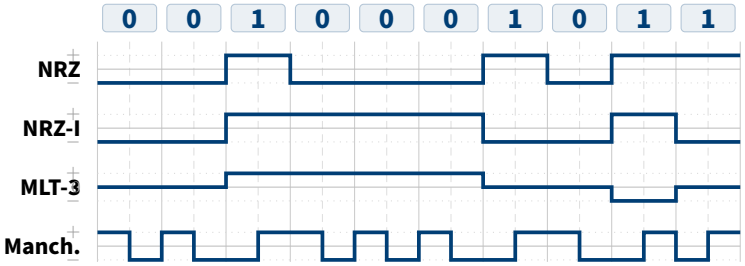
Zeichnen Sie für die folgenden Bitmuster jeweils die Signalverläufe in die vorbereiteten Diagramme ein:
NRZ: $0 = -, 1 = +$ | **NRZ-I:** $1 =$ Pegelwechsel (Start mit $-$) | **MLT-3:** $1 =$ Pegelübergang im Zyklus $0 \to + \to 0 \to - \to 0$ (Start mit 0, nächster Pegel bei 1 ist $+$) | **Manchester (IEEE 802.3):** $0 =$ Flanke $+ \to -$ in Bitmitte (↓), $1 =$ Flanke $- \to +$ in Bitmitte (↑).

1.) Bitmuster: `1001001101`



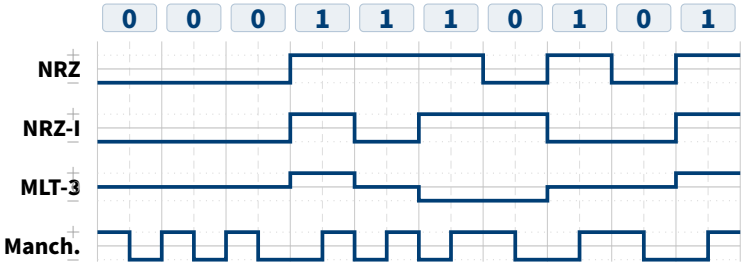
Simulator ↗
Signalplotter

2.) Bitmuster: `0010001011`



Simulator ↗
Signalplotter

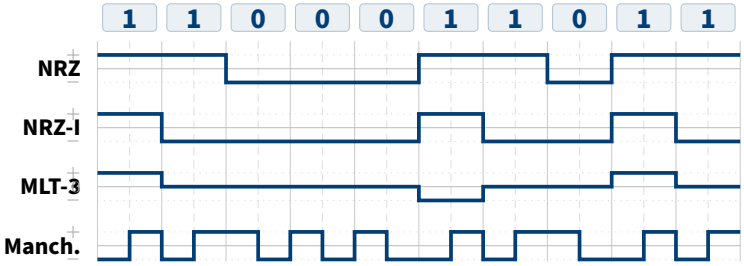
3.) Bitmuster: `0001110101`



Simulator ↗

Signalplotter

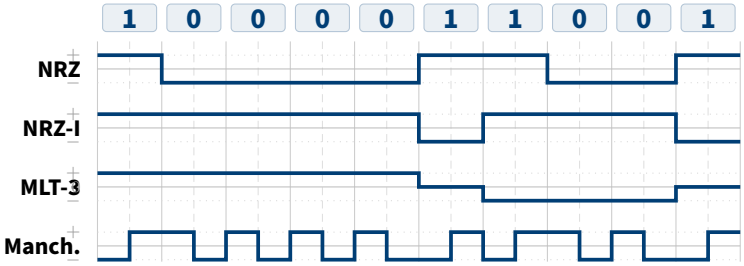
4.) Bitmuster: `1100011011`



Simulator ↗

Signalplotter

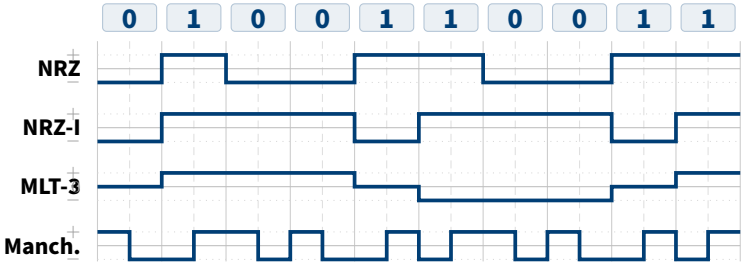
5.) Bitmuster: `1000011001`



Simulator

Signalplotter

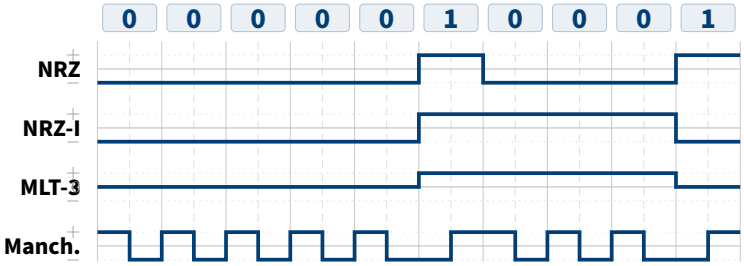
6.) Bitmuster: `0100110011`



Simulator

Signalplotter

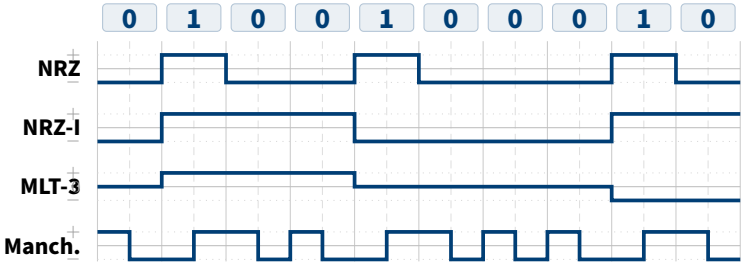
7.) Bitmuster: `0000010001`



Simulator ↗

Signalplotter

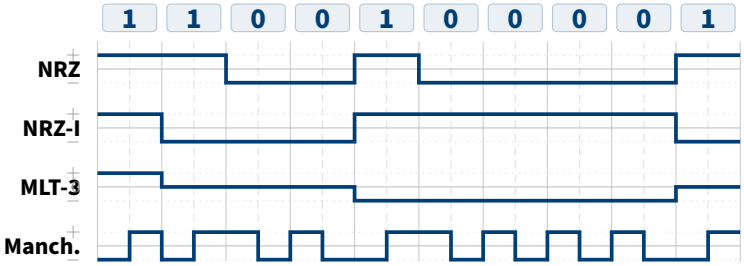
8.) Bitmuster: `0100100010`



Simulator ↗

Signalplotter

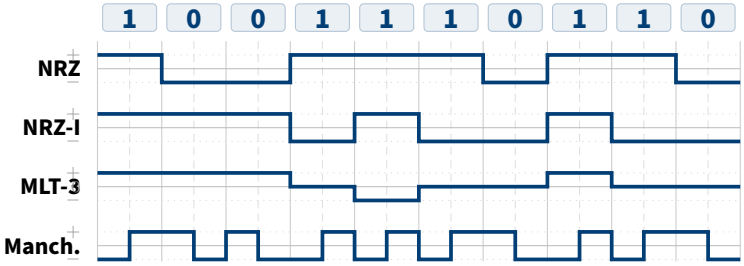
9.) Bitmuster: `1100100001`



Simulator ↗

Signalplotter

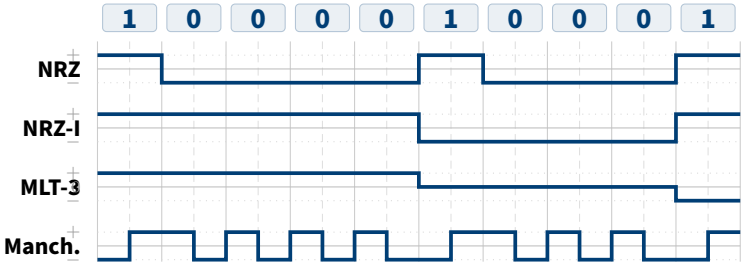
10.) Bitmuster: `1001110110`



Simulator ↗

Signalplotter

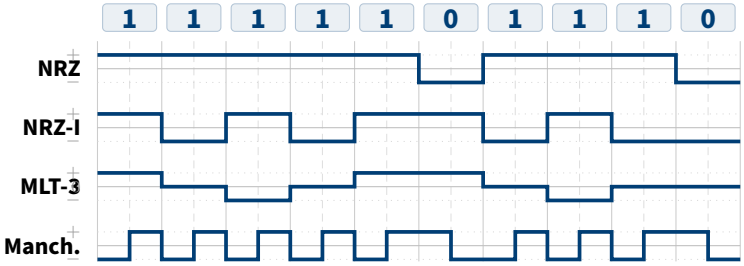
11.) Bitmuster: `1000010001`



Simulator

Signalplotter

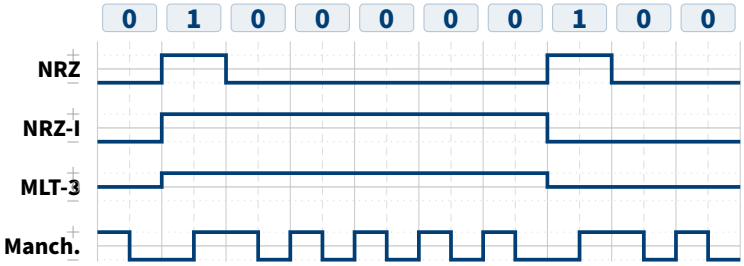
12.) Bitmuster: `1111101110`



Simulator

Signalplotter

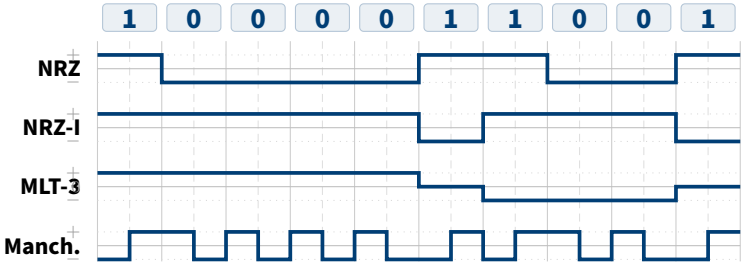
13.) Bitmuster: `0100000100`



Simulator

Signalplotter

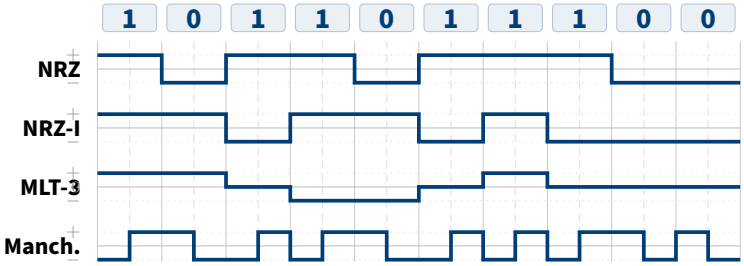
14.) Bitmuster: `1000011001`



Simulator

Signalplotter

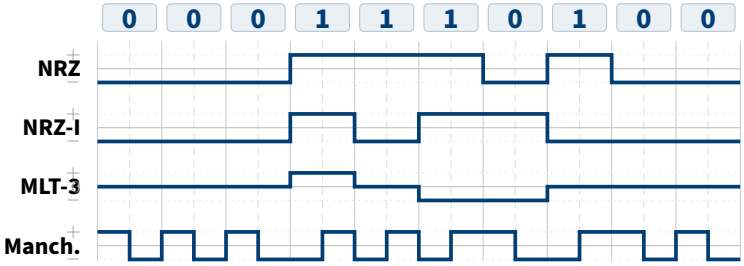
15.) Bitmuster: `1011011100`



Simulator ↗

Signalplotter

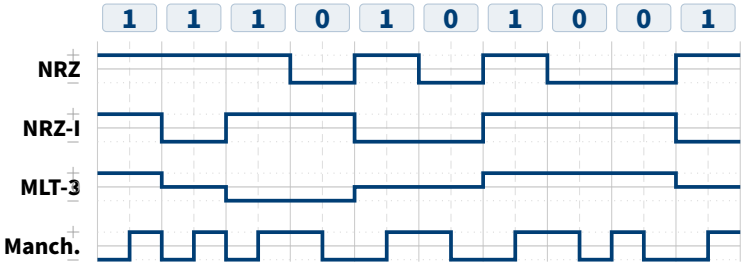
16.) Bitmuster: `0001110100`



Simulator ↗

Signalplotter

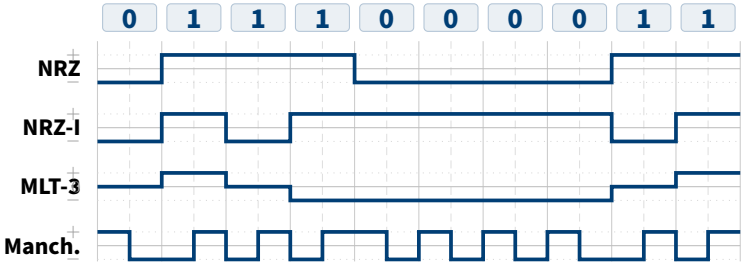
17.) Bitmuster: `1110101001`



Simulator ↗

Signalplotter

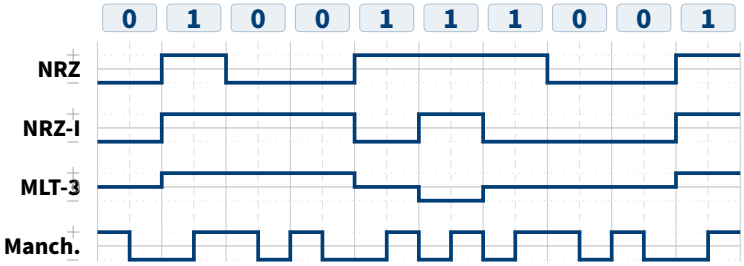
18.) Bitmuster: `0111000011`



Simulator ↗

Signalplotter

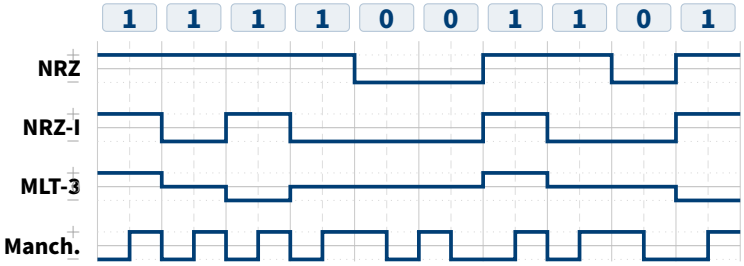
19.) Bitmuster: `0100111001`



Simulator ↗

Signalplotter

20.) Bitmuster: `1111001101`



Simulator ↗

Signalplotter